Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/23/2007
PI Name:	Albert, Mitchell S Ph.D.		
Project Title:	Hyperpolarized 129Xe MRI of the Brain, Lungs and Plants Augmented with MRI Pulse Sequence Development		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01655	**Congressional District:**	3
Comments:	Note: PI moved to U Mass Medical School in 2007; previously at Brigham and Women's Hospital/Harvard University (9/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	PECASE AWARD
Start Date:	09/30/2004	**End Date:**	01/31/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed back to 1/31/2007 per J. Dardano/JSC, as NCE referred to same project with new grant number; also chgd start date per info fr JSC (info rec'd 9/2008) NOTE: Received no-cost extension to 1/31/2008 per P. Denkins/JSC (3/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reilly, John Joseph (Brigham and Women's Hospital) Mulkern, Robert (Children's Hospital Boston)		
Grant/Contract No.:	NAG9-1041		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We are developing a new diagnostic technology: magnetic resonance imaging (MRI) and spectroscopy of hyperpolarized noble gases. Large non-equilibrium nuclear spin polarizations can be created in the spin-1/2 noble gases, 3He and 129Xe, using the technique of spin-exchange optical pumping. Such large polarizations greatly enhance the magnetic resonance detection of 3He and 129Xe, enabling high-resolution MRI studies of the lung and airways, as well as dissolved tissue-phase investigations of organs and tissues, such as the brain, using lipophilic 129Xe. An important feature of hyperpolarized noble gas MR is that it does not require a large magnetic field, unlike the conventional water proton MR method used in medical MRI. Low magnetic field MRI with hyperpolarized noble gases can allow us to use a small, light-weight, low-power devices. This capability provides potential for advances in both portable ground-based and practical space-based MR systems. Our proposed ground-based investigations with hyperpolarized noble gas MR will affect biomedical research in two important ways: (1) by providing the potential to acquire novel information about physiological structure and function of the lung and the brain in space-flight and in terrestrial gravity, and (2) by developing a new diagnostic technology that has clinical applications for diagnosis of pathology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An important feature of hyperpolarized noble gas MR is that it does not require a large magnetic field, unlike the conventional thermally polarized water proton MR method used in medical MRI. As a result, at low magnetic fields (0.01 Tesla) the signal-to-noise ratio (SNR) of hyperpolarized noble gas MR is much larger than the SNR of conventional thermally polarized MR. Low field MR utilizing hyperpolarized noble gases should be practical in a small, light-weight, low-power device, providing the potential for both portable ground-based and practical space-based MR systems. Specific magnetic and biological properties of hyperpolarized 3He and 129Xe MRI include: (i) airway and lung gas space imaging (for diagnosis of chronic obstructive pulmonary disease (COPD), e.g. emphysema, asthma, and for studies of respiratory function); (ii) the imaging of lipid structures in the brain (for the diagnosis and staging of demyelinating diseases such as multiple sclerosis, and in research on brain function); and (iii) angiography and measurement of blood flow (for diagnosis of stroke and ischemia, and for research into the function of the brain and cardiopulmonary system). The recent progress made in our lab lays the foundation for the continued development of hyperpolarized xenon MRI for brain imaging. Specifically, imaging of lipid structures in the brain (for the diagnosis and staging of demyelinating diseases such as multiple sclerosis) and angiography and measurement of blood flow (for diagnosis of stroke and ischemia, and for research into the function of the brain and cardiopulmonary system), are now viable research goals. Because HP 129Xe is an MR signal source with properties very different from those generated from water-protons, it may yield structural and functional information not detectable by proton-based MRI methods. Thus brain imaging with hyperpolarized xenon should be of great benefit to the diagnosis and treatment monitoring of a number of neurological diseases.
Task Progress:	The overall goal of this proposal is to develop hyperpolarized noble gas magnetic resonance imaging (HP MRI) for the assessment of pulmonary and nervous system damage that may be incurred during human space flight. Hyperpolarized noble gas MRI is showing dramatic results for diagnostic imaging of the lungs, brain, and other parts of the body, some of which have been resistant to conventional proton MRI. Most notably, we have achieved highresolution dynamic images of the lungs and airways using hyperpolarized 3He, and have measured cerebral perfusion and functional activity in the brain using hyperpolarized 129Xe (HP 129Xe). The number of space flight-related pathologies which may benefit from this technique are vast and range from pulmonary dysfunction caused by microgravity to cognitive and neurobehavioral problems resulting from the stress and isolation of extended term space travel. This novel technology should also offer a rapid, sensitive method to detect treatment effects and thus may expedite the development of therapeutic strategies for treating space flight related injury, such as aerosol administration of drugs via the lungs, which may obviate the need for medicinal injections during space flight. [editor's note: Project continues with grant number NNX07AP80G; see that project for FY08 forward (9/08)]
Bibliography Type:	Description: (Last Updated: 03/23/2012)
Abstracts for Journals and Proceedings	Tzeng Y-S, Shah NM, Mansour JK, Gereige J, Washko GR, Cho MH, Stepp E, Lutchen KR, Albert MS. "Response to Deep Inspirations in Reactive and Non-Reactive Asthmatic Subjects Using Hyperpolarized 3He MRI." To be presented at the 15th Annual Meeting ISMRM, Berlin, Germany, May 2007. Proceedings of ISMRM 15th Annual Meeting and Exhibition, May 2007. , May-2007
Abstracts for Journals and Proceedings	Tzeng Y-S, Gereige J, Szender JB, Zhalehdoust-Sani S, Washko GR, Cho MH, Shah NM, Mansour JK, Stepp E, Albert MS, Lutchen KR. "Similar Responses to Deep Inspirations Observed in Reactive and Non-Reactive Asthmatic Subjects Using Hyperpolarized 3He MRI." To be presented at the American Thoracic Society Meeting 2007, San Francisco, May 2007. Proceedings of the American Thoracic Society 2007. , May-2007